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The Role of Mineralogy and Physicochemical Conditions on Geological Hydrogen (H₂) Generation in Basalts and Ultramafic Rocks

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Abstract

Hydrogen (H₂) generation during serpentinization has long been explored, with traditional mechanisms centered on three key processes: the hydration of olivine and pyroxenes during serpentine formation, brucite precipitation from Mg-rich mineral reactions, and the oxidation of ferrous (Fe²⁺) to ferric (Fe³⁺) iron, often accompanied by the formation of magnetite. These reactions are known to overlap, reinforcing the view that mineralogical composition and physicochemical conditions critically shape H₂ evolution in natural systems.

This study investigates the impact of mineralogy on H₂ generation in ultramafic and mafic (basaltic) rocks under controlled hydrothermal conditions. Seven rock types were examined in laboratory experiments conducted at 150 °C, 500 psi, and an initial pH of 8.5. Hydrogen was measured using an electrochemical analyzer, while fluid composition was assessed using Inductively Coupled Plasma Mass Spectrometry (ICP-MS). Post-reaction solids were analyzed via X-ray Diffraction (XRD) to identify mineral transformations.

Results reveal that both ultramafic and basaltic rocks can generate H₂, but with widely varying yields that do not correlate directly with bulk Fe content. Instead, differences in mineralogy, trace element composition, and fluid–rock interactions strongly influence hydrogen output. Importantly, the study finds that H₂ generation does not always require the formation of serpentine, brucite, or magnetite. Instead, alternate pathways involving Fe-rich silicates, aluminosilicates, and transition metal-bearing minerals may promote H₂ production through surface-mediated redox reactions and geocatalytic effects.

These findings broaden the understanding of natural H₂ generation beyond classic serpentinization and highlight the role of mineral surface chemistry and trace elements in sustaining hydrogen release. Such insights could inform future strategies for geologic hydrogen exploration and the design of engineered subsurface systems for low-carbon H₂ production.

Keywords: Geologic hydrogen, Physicochemical conditions, Serpentinization, Ultramafic and basalt rocks, Mineralogy

Introduction

Subsurface hydrogen (H₂) generation has become a topic of significant scientific and technological interest due to its relevance to early Earth biogeochemistry, modern microbial ecosystems, and the growing need for

low-carbon, sustainable energy resources. Among the dominant geological processes generating hydrogen is the interaction between water and iron (Fe(II))-bearing silicate minerals, particularly those present in mafic and ultramafic rocks. Despite the growing attention to serpentinization and related mineral-fluid reactions, the underlying mechanisms responsible for hydrogen production remain poorly resolved.

Recent estimates highlight the significance of continental lithospheric contributions to natural hydrogen generation, with Precambrian cratonic rocks alone estimated to produce between 0.36 and 2.27×10^{11} moles of H_2 annually (Lollar et al., 2014). This flux is comparable to, and possibly even exceeds, the hydrogen production associated with oceanic lithosphere and mid-ocean ridge environments. The high production potential is primarily linked to reactions involving olivine and pyroxene-rich rocks, which dominate the ultramafic and mafic crust. Among these reactions, serpentinization, a hydration process of olivine that forms serpentine and magnetite, has historically received the most attention. However, other mineral-fluid interaction pathways such as iron oxidation, silicate hydrolysis, and coupled redox reactions involving transition metals are increasingly recognized as significant contributors to H_2 generation (Mayhew et al., 2013; Mohammed et al., 2024; Templeton et al., 2024).

Beyond geochemistry, subsurface H_2 plays a pivotal role in supporting microbial life in the deep biosphere, where sunlight is absent and energy must be derived from geochemical sources. Additionally, the study of geological H_2 generation has relevance for understanding the conditions that prevailed during the early Earth. During the Archean Eon, serpentinization and related water-rock reactions were likely more widespread due to higher geothermal gradients and the presence of more abundant ultramafic rocks at or near the surface. These reactions are believed to have contributed to prebiotic chemistry and the emergence of life by providing reducing conditions, molecular hydrogen, and other geochemical gradients necessary for early metabolic pathways (Schulte et al., 2006; Nisbet & Sleep, 2001). As such, studying H_2 generation in modern analog systems can provide critical clues to how life may have originated and evolved.

From a technological perspective, natural hydrogen, sometimes referred to as "white" or "geologic" hydrogen, is increasingly being explored as a potential renewable energy resource. In contrast to hydrogen derived from fossil fuels or electrolysis (i.e., grey, blue, or green hydrogen), natural H_2 production offers a passive, low-carbon alternative that leverages Earth's own geological processes. Harnessing this energy source, however, requires a detailed understanding of the mineralogical and geochemical parameters that control its formation, accumulation, and migration (Mayhew et al., 2013; Corre et al., 2023; Templeton et al., 2024; Ali et al., 2025).

Despite three decades of progress in understanding serpentinization and hydrothermal alteration processes, several fundamental questions remain unanswered. These include uncertainties in the reaction kinetics governing H_2 release, the identity and distribution of rate-limiting steps, and the role of catalytic effects from trace elements and mineral surfaces. For instance, while Fe(II) oxidation is a key reaction, the mechanisms by which transition metals, such as Ni, Mn, Cu, or Mo, mediate redox cycling and enhance H_2 production are not well understood. Similarly, it remains unclear why rocks with similar bulk Fe content exhibit stark differences in hydrogen output under comparable experimental conditions, suggesting a critical role for factors such as crystal structure, surface area, mineral grain size, and alteration textures.

To address these gaps, this study presents an integrative experimental and geochemical investigation into H_2 production from a set of seven mineralogically diverse mafic and ultramafic rocks. By combining laboratory-based hydrothermal experiments with detailed mineralogical, chemical, and surface characterization, we seek to clarify the interplay between rock composition, fluid chemistry, and environmental conditions in controlling H_2 generation.

The experimental setup involved exposing powdered rock samples to slightly alkaline water (pH 8.5) at elevated temperature (150 °C) and pressure (500 psi), using an inert nitrogen (N_2) atmosphere to simulate subsurface conditions. At the end of the experiments, fluid samples were analyzed for dissolved H_2 and ionic species, while post-reaction solids were subjected to X-ray Diffraction (XRD) analyses to monitor mineral

transformations. These analyses enabled the correlation of H₂ production rates with specific mineralogical features and the formation of secondary phases.

The study's primary objectives are:

1. To quantify baseline hydrogen production rates from distinct mafic and ultramafic lithologies;
2. To identify the mineral phases and geochemical environments most conducive to sustained hydrogen generation.

Ultimately, this work aims to move beyond a purely serpentinization-centered framework by highlighting alternative reaction possibilities and identifying the previously overlooked impact of trace minerals. These insights contribute not only to our fundamental understanding of subsurface geochemistry but also to the growing field of natural hydrogen exploration, with potential implications for sustainable energy development.

Methodology

Elemental and Mineralogical Characterization

Seven rock samples, labeled A through G, were evaluated for their potential to generate hydrogen (H₂) under controlled hydrothermal laboratory conditions. Five of the samples were ultramafic in composition, while two were classified as mafic (basaltic). Among the ultramafic group were two dunite specimens from the Trinity Ophiolite (Samples A and E), along with three peridotite samples. Sample B was sourced from Ward's Science, whereas Samples C and D were obtained from New England Research. The mafic group consisted of Samples F and G, both basaltic in nature and collected from sites in the United Kingdom, representing a geographically diverse set of materials.

X-ray fluorescence (XRF) analysis was used to determine the elemental composition of each sample, as shown in Fig. 1. While silicon (Si) and iron (Fe) contents were generally similar between the two rock groups, distinct differences were observed in other major elements. The mafic rocks (Samples F and G) were notably enriched in aluminum (Al), calcium (Ca), sodium (Na), and potassium (K), but had significantly lower concentrations of magnesium (Mg) compared to the ultramafic rocks (Samples A through E). These compositional differences reflect underlying mineralogical and geochemical contrasts, which are likely to influence the capacity of each rock type to produce hydrogen under hydrothermal conditions.

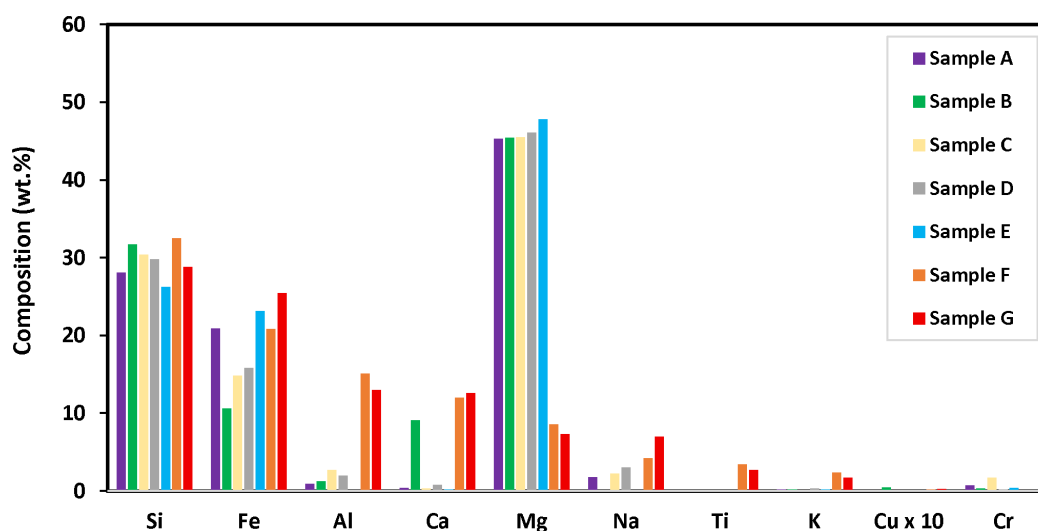


Figure 1—Elemental composition of the unreacted rock samples using X-Ray fluorescence (XRF) analysis: Samples A to E are ultramafic, and samples F and G are mafic.

The samples encompass a variety of peridotite compositions with differing levels of serpentinization and mineralogical characteristics. The mafic rocks, in contrast to their ultramafic counterparts, show distinct differences in both mineral composition and alteration history. While the ultramafic samples are generally dominated by olivine and pyroxenes, the basaltic rocks are characterized by significant amounts of plagioclase feldspars, phyllosilicates, and pyroxene minerals that may influence H₂ generation in different ways.

Experimental Description

In the experiments, rock samples were crushed using a porcelain mortar and pestle to prevent metal contamination, and then sieved to obtain particles between 150 and 300 μm . The hydrothermal reactions were conducted over a 7-day period using a water-to-rock ratio of 2:1, an initial pH of 8.5, at 150°C, and 500 psi. The mixtures were loaded into a 150 mL Hastelloy mini-autoclave, purged with high-purity nitrogen (99.99%) to eliminate air, and pressurized to 320 psi. The autoclave was then heated at a rate of 10°C/min to 150°C ($\pm 1^\circ\text{C}$), causing the internal pressure to rise to 500 psi. These conditions replicate subsurface hydrothermal environments typical of serpentinized systems, where natural hydrogen is generated through interactions between rock and water.

Post-Reaction Fluids and Solids Characterization

Comprehensive characterization of the reaction products was conducted to evaluate the extent of H₂ generation and assess the geochemical and mineralogical changes resulting from rock–fluid interactions. After the experiment, the system was allowed to cool to ambient temperature for one hour before collecting the generated hydrogen gas. The H₂ was captured in a sealed, transparent plastic bag and analyzed using an electrochemical gas analyzer (accuracy ± 2 ppm).

The aqueous supernatant was sampled and subjected to Inductively Coupled Plasma Mass Spectrometry (ICP-MS) to determine the elemental composition, including trace elements mobilized during the reaction. Additionally, physicochemical parameters such as total dissolved solids (TDS), conductivity, and pH were measured using a multiparameter meter to track the evolution of the fluid chemistry. These measurements served as indicators of ongoing dissolution–precipitation reactions and ion exchange processes, with TDS and conductivity reflecting ionic strength and mobility, while pH indicated acid-base shifts within the system.

To evaluate changes in the mineralogy of the solid phases, X-ray Diffraction (XRD) analysis was performed. Unreacted and reacted rock samples were oven-dried at 60 °C overnight, homogenized, and analyzed using a Malvern Empyrean PANalytical diffractometer (Cu K α radiation, 45 kV, 40 mA, scan range 4.5°–70° 2 θ). Phase identification and quantification were conducted using the HighScore Plus software integrated with the ICDD PDF-4-2025 database. This allowed detection of primary mineral dissolution, secondary mineral formation, and other phase transformations induced by hydrothermal reactions.

Results and Discussion

Hydrogen Generation

As illustrated in Fig. 2, both ultramafic and mafic rocks exhibit a wide range of H₂ production levels. For instance, among the ultramafic samples, Sample C generated as little as 1.1 mmol/kg of rock, while Sample B reached up to 9.6 mmol/kg, highlighting substantial variability even within the same rock category. A similar pattern is observed in mafic rocks, where Sample F produced 10.8 mmol/kg, compared to 0.76 mmol/kg from Sample G. These differences underscore the non-uniform nature of H₂ generation across both rock types, emphasizing the influence of mineralogical and geochemical properties on H₂ production during serpentinization and related hydro-geochemical processes.

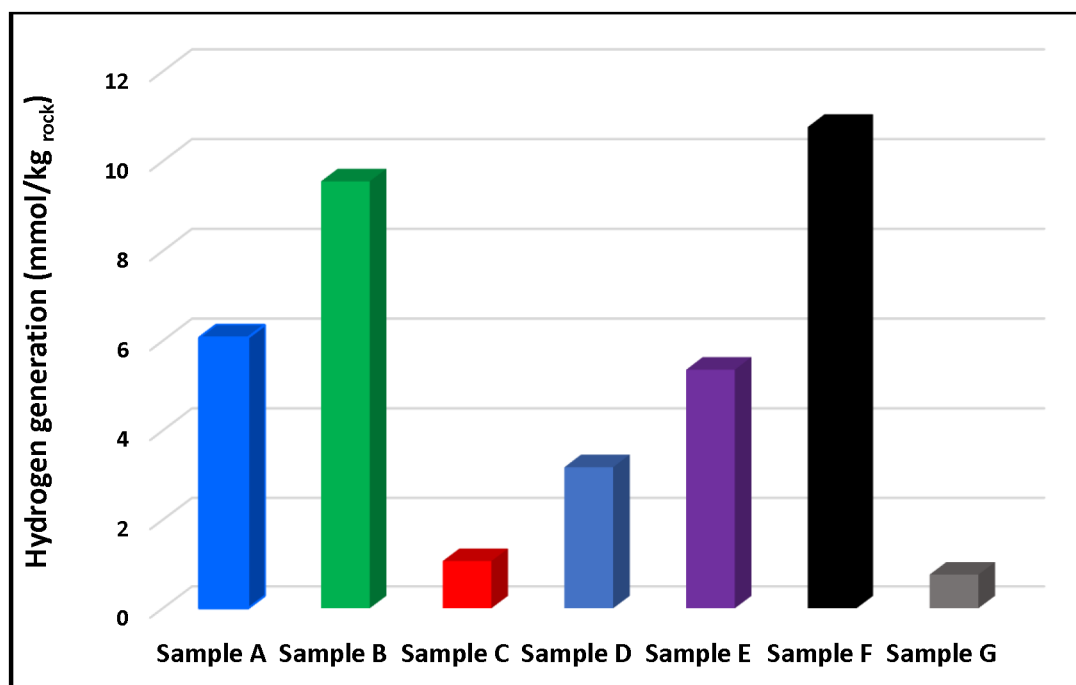


Figure 2—Hydrogen generation recorded for different ultramafic (dashed lines; Samples A to E) and mafic (solid lines; Samples E and F) rocks after 7 days of hydrothermal exposure, at 150 °C, pressure of 500 psi, and fluid initial pH of 8.5.

Results indicated that H_2 generation in ultramafic rock systems is a complex process, with distinct H_2 production for various rocks under the same hydrothermal conditions. The hydrothermal reactions of different rock types exhibit distinct reactivity patterns. Even trace mineral components in the bulk rock composition are likely to significantly influence overall reaction kinetics and H_2 production. Rather than being strictly governed by iron oxidation states, reaction rates appear to depend on mineralogical composition and kinetics. The preliminary data indicate that both bulk mineralogy and elemental-scale transformations may contribute to the variability in H_2 generation across different rock types. Although H_2 generation is linked to the oxidation of ferrous iron in rocks, the hydrous alteration of igneous rocks with similar ferrous iron content does not always result in equivalent H_2 amounts.

Interestingly, Fe-rich mineral phases such as olivine and clinopyroxene (enstatite) were identified in Sample F. This mineral transformation was accompanied by the consumption of cristobalite (SiO_2), a silica polymorph that may signify accelerated reaction kinetics. The detection of these Fe-rich phases is particularly important, as iron oxidation and redox cycling are well-established mechanisms that contribute to H_2 generation (Marcaillou et al., 2011; McCollom, 2020, 2022). These findings suggest that the increased H_2 production observed in Sample F is primarily driven by active Fe-mediated redox processes and the emergence of reactive mineral surfaces. This may explain why this mafic rock exhibited the highest H_2 output, even surpassing that of all the ultramafic rocks examined. Conversely, another mafic rock, Sample G, showed the lowest H_2 production among all samples studied. This low H_2 yield may be attributed to its comparatively lower initial abundance of Fe-rich minerals in the unreacted state, as illustrated in Fig. 3.

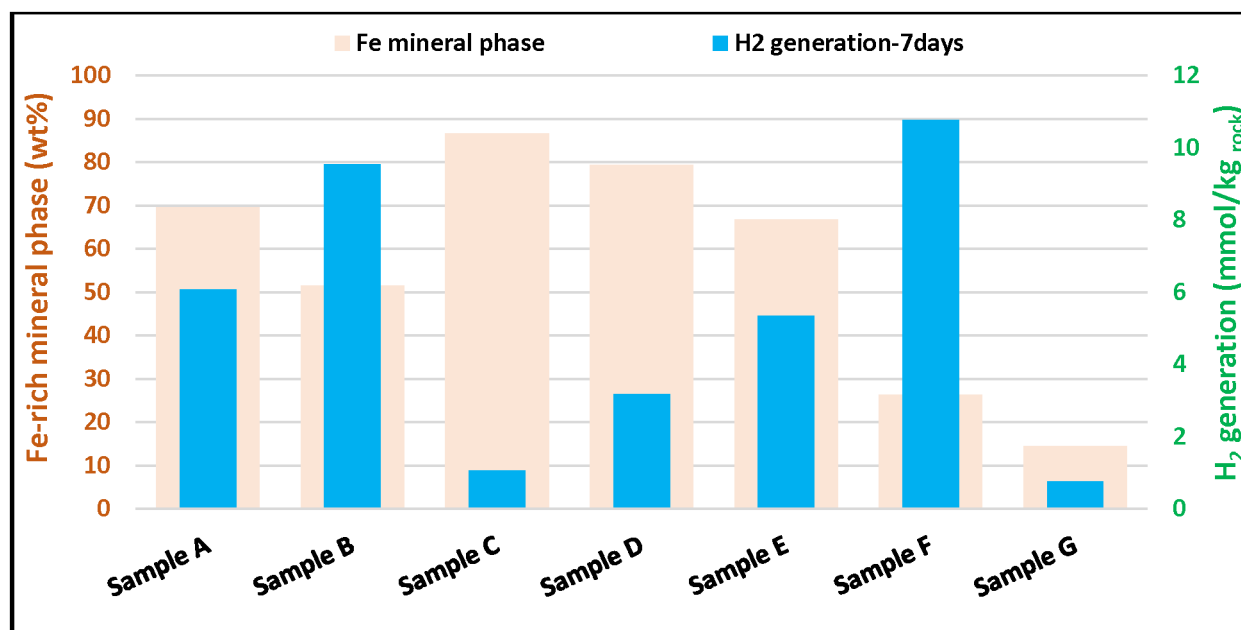


Figure 3—Iron-rich mineral phase against H₂ generation for the ultramafic and basaltic rocks after 7 days of hydrothermal exposure, at 150 °C, pressure of 500 psi, and fluid initial pH of 8.5.

Minerals Transformation and H₂ Generation in Ultramafic Rocks

The mineralogical composition of ultramafic rocks varied significantly with reaction time, highlighting distinct alteration trends linked to H₂ generation. In Sample A, olivine content progressively decreased from 59.5 wt.% to 55.3 wt.% over 7 days, with minimal brucite formation during the reaction (Fig. 4). Sample B exhibited a marked transformation, with lizardite increasing from 38.4 wt.% to 68.5 wt.% over 7 days, accompanied by reductions in olivine, hematite, and clinopyroxene, consistent with serpentinization reactions known to produce H₂. A decline in birnessite, a manganese hydroxide mineral, also suggested additional geochemical activity.

Sample C followed a similar pattern, with olivine decreasing from 82.4 wt.% to 74.6 wt.%, and corresponding increases in lizardite and brucite content. Unlike Sample A, hematite content in Sample C increased over time, though magnetite remained absent. Overall, the XRD results indicated that Fe-rich minerals, such as olivine, hematite, and magnetite, actively participated in the transformation processes associated with H₂ production.

Samples D and E demonstrated appreciable H₂ generation of 3.2 and 5.4 mmol/kg of rock, respectively (see Fig. 3), despite the absence of detectable magnetite in their post-reaction XRD patterns. In Sample D, olivine content declined from 70.7 wt.% to 62.8 wt.% over 7 days, while lizardite increased to 23.9 wt.% from an initial 15.2 wt.%, and brucite appeared at 1.4 wt.%, consistent with serpentinization reactions. These processes involve olivine hydration and Fe²⁺ oxidation from wüstite, resulting in H₂ release. Sample E showed even higher H₂ production and a significant increase in lizardite content from 19.6 wt.% to 36.7 wt.% without the formation of brucite or magnetite. This suggests that H₂ production can occur through alternative pathways beyond classic mineral transformation. The data imply a potential role of surface geocatalysis, where mineral surfaces may facilitate redox reactions that sustain H₂ generation even after major phase transitions have occurred.

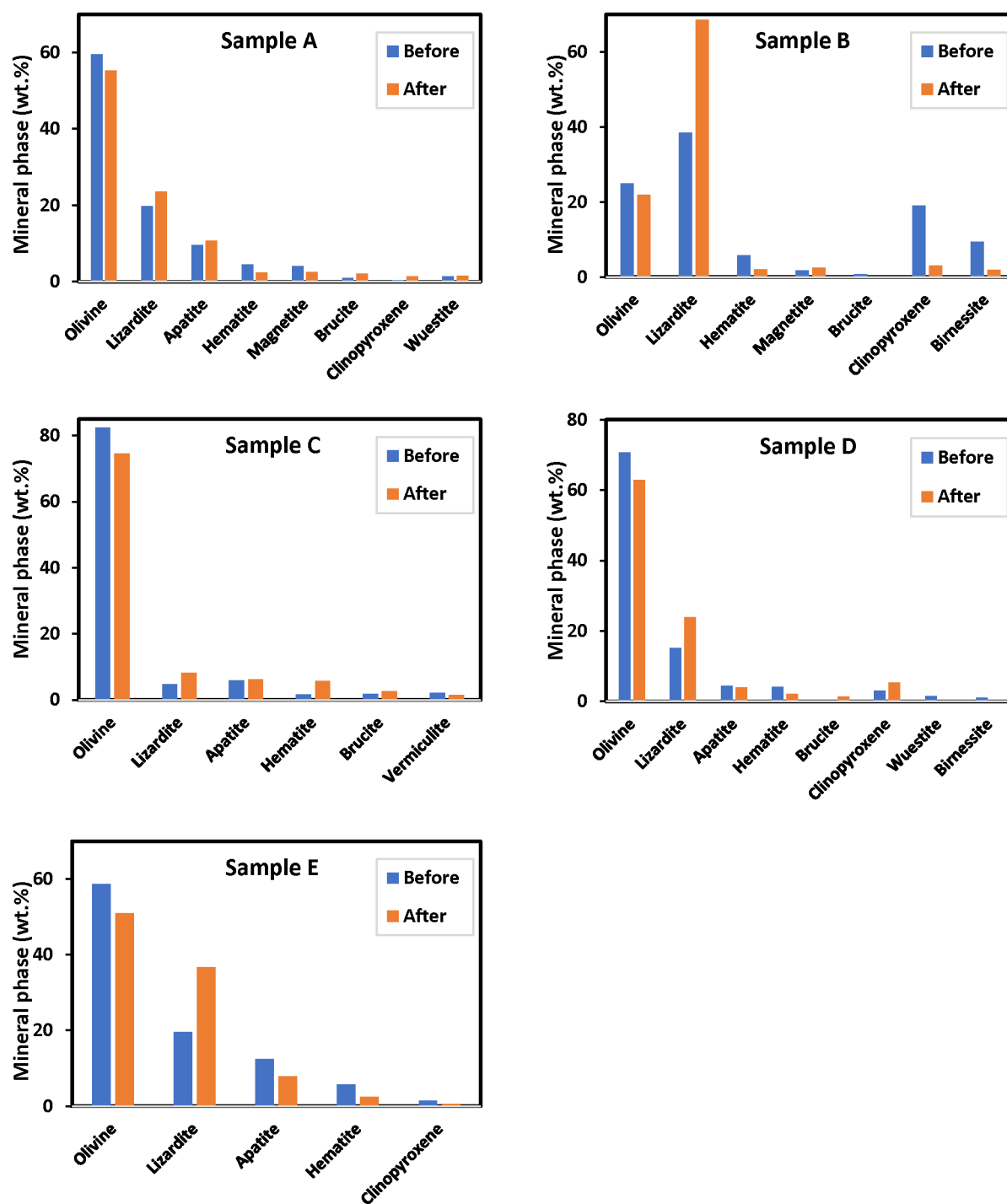


Figure 4—XRD mineralogical transformation of the mafic ultramafic rock after 7 days of hydrothermal interaction.

Minerals Transformation and H₂ Generation in Mafic (Basaltic) Rocks

The studied mafic rocks were mainly composed of plagioclase minerals, particularly labradorite, andesine, and anorthite alongside minor olivine and trace silicates such as biotite, orthopyroxene (enstatite), clinopyroxene (diopside), and zeolite as illustrated in Fig. 5. Although Samples F and G shared broadly similar initial mineral compositions, they displayed vastly different H₂ production levels. Sample F generated significantly more H₂ (10.8 mmol/kg) compared to Sample G (0.76 mmol/kg), implying that factors beyond bulk mineral content, such as reaction pathways and mineralogical transformations, may be critical to H₂ generation.

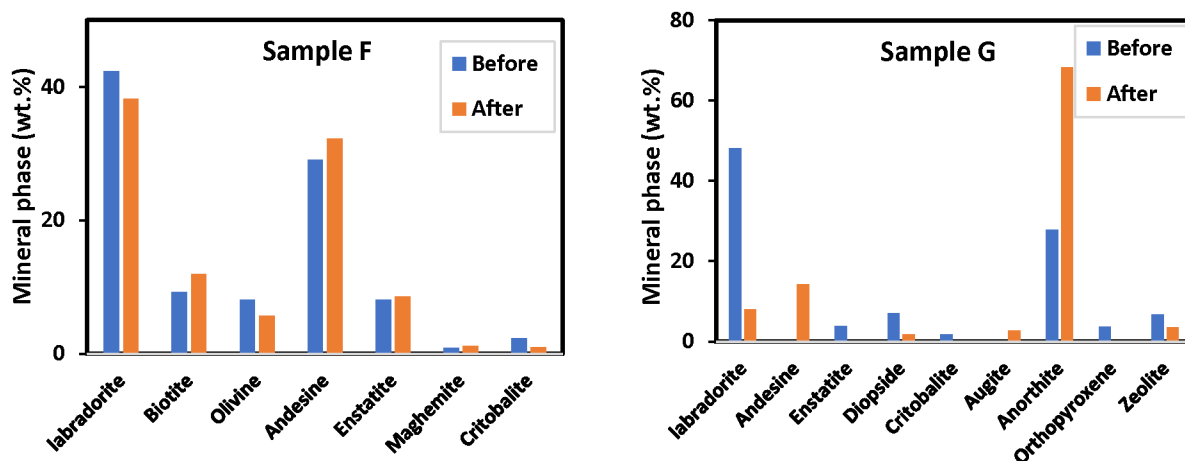


Figure 5—XRD mineralogical transformation of the mafic rock after 7 days of hydrothermal interaction.

Post-reaction analysis revealed divergent geochemical behavior: in Sample F, parts of the labradorite transformed into andesine, suggesting sodium incorporation or calcium loss, whereas in Sample G, transformation to anorthite indicated sodium depletion or calcium enrichment. These shifts in elemental distribution point to distinct mineral alteration mechanisms that may underlie the difference in H_2 output between the two basaltic samples.

Geochemical Evolution of Aqueous Fluids and Its Role in H_2 Generation

Elemental analysis of the aqueous fluids (via ICP-MS) following rock-water reactions revealed that both dissolution and precipitation processes played key roles in altering fluid chemistry and potentially influencing H_2 generation. Distinct fluid-rock interactions were observed across ultramafic and mafic rock types, indicating that different geochemical pathways contribute to H_2 production.

Among the ultramafic rocks, Sample A showed elevated Mg and Ca concentrations in solution, implying dissolution of Mg- and Ca-bearing minerals, while Sample B exhibited minimal fluid alteration, possibly due to precipitation of Mg and Zn into secondary phases. Sample C showed marked magnesium enrichment, indicating extensive dissolution of Mg-rich phases, such as olivine. In Sample D, potassium release suggested the breakdown of K-bearing phyllosilicates, and in Sample E, increased Calcium levels indicated the dissolution of Ca-rich minerals, such as clinopyroxenes.

In mafic rocks, fluid compositions also evolved differently. Sample G exhibited significant increases in K, Ca, and Zn, suggesting dissolution of feldspars and calcic silicates like labradorite and clinopyroxene. In contrast, Sample F primarily showed elevated K, likely from the dissolution of biotite or other K-rich micas. Notably, no dissolved silica (Si) was detected in any sample, despite its known role in suppressing H_2 production. This highlights that other solutes and elemental interactions may dominate in these systems (Leong et al., 2021; Tutolo et al., 2020).

Overall, the variability in elemental mobility across samples reflects differences in mineralogy, solubility, and fluid-rock interaction kinetics. These processes collectively influence the extent and mechanism of H_2 production, extending beyond Fe oxidation to include broader mineral transformation dynamics.

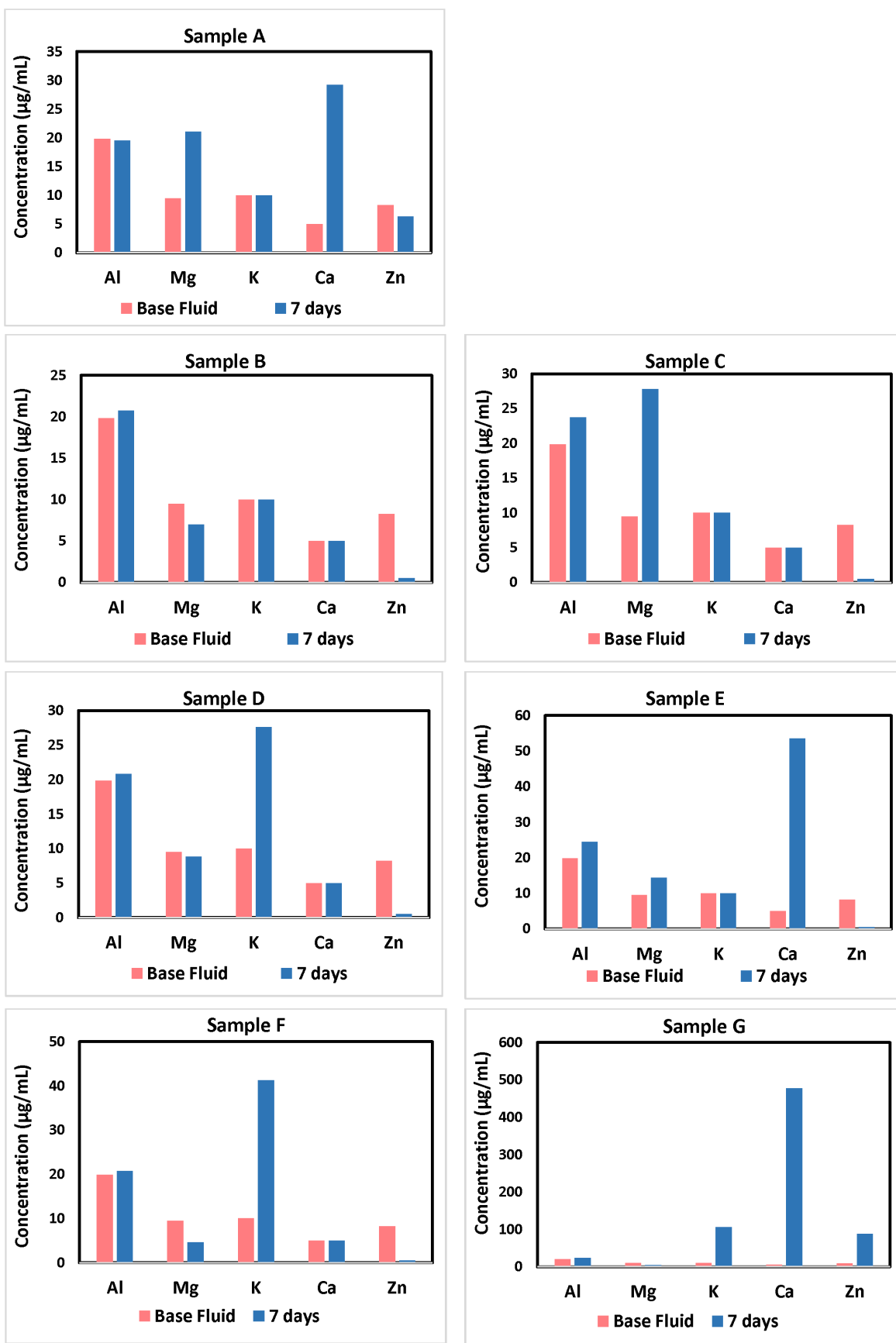


Figure 6—The composition of the major elements in the aqueous solution, including Al, Mg, K, Ca, and Zn, was analyzed to assess the geochemical evolution of the aqueous fluids.

Physicochemical Properties of Reaction Fluids and Implications for H₂ Generation

Analysis of total dissolved solids (TDS), electrical conductivity, and pH provided insights into the physicochemical evolution of reaction fluids and their implications for H₂ generation, this is presented in Fig. 7. TDS and conductivity reflect the ionic strength of the fluid, driven primarily by mineral dissolution processes that release ions such as Mg²⁺, Ca²⁺, K⁺, and Na⁺. Increases in these parameters typically indicate enhanced dissolution, while decreases suggest ion removal through precipitation into secondary mineral phases.

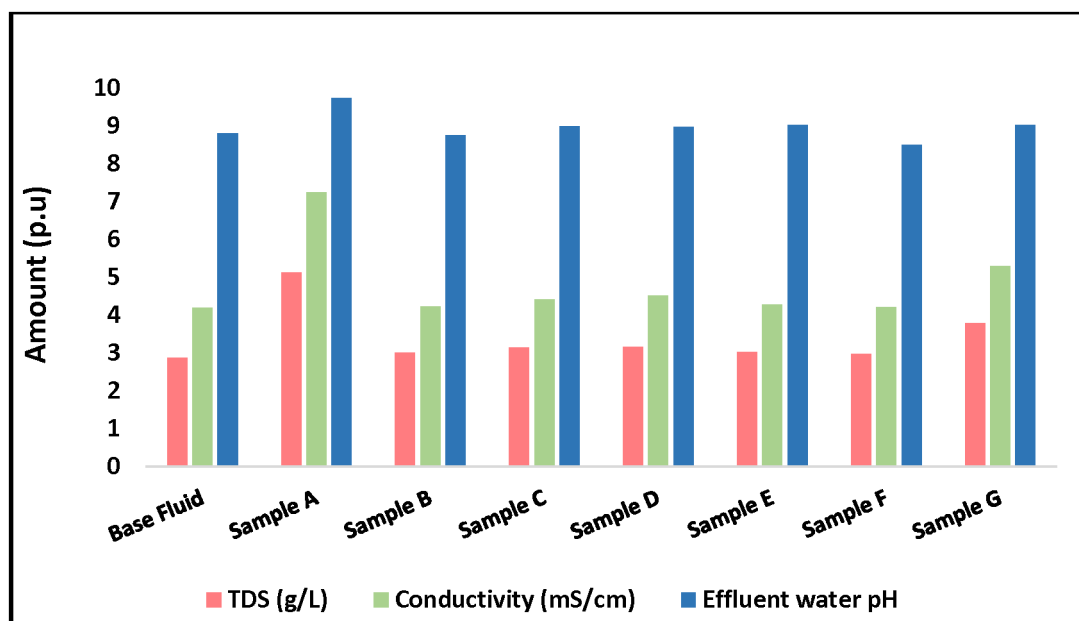


Figure 7—The total dissolved solids (TDS), conductivity, and pH of the reaction fluids to assess the physicochemical changes of the aqueous fluids.

The fluids' pH trends offer additional insight into geochemical reactions. Elevated pH levels in most of the ultramafic (except Sample B) and mafic (Sample G) rocks suggest hydroxide formation or proton consumption during mineral reactions. However, the relationship between these physicochemical properties and H₂ production is non-linear. For example, although both Samples A and C showed increased pH, their H₂ yields differed significantly; Sample A produced high H₂, while Sample C generated the least among ultramafic rocks. A similar pattern was observed among mafic samples, where Sample G exhibited higher TDS, conductivity, and pH but produced less H₂ than Sample F.

Geochemical Influence of Transition Metals on H₂ Generation

Fig. 8 shows the relationship between hydrogen generation and the X-ray fluorescence (XRF) percentage of three trace elements: titanium (Ti), copper (Cu), and chromium (Cr) across the seven rock samples. In Fig. 8, the percentage of copper is multiplied by 10 to enable it to be plotted on the same scale as titanium and chromium. For the ultramafic rocks (Samples A to E), a positive trend is observed between hydrogen generation and both Ti and Cu concentrations. Sample B, which displays relatively elevated Ti and Cu content, yields one of the highest hydrogen generation values among the ultramafic group. Likewise, Samples A and E, with modest Ti and Cu concentrations, also produce considerable hydrogen, reinforcing this positive correlation. Conversely, Cr appears to exhibit a negative relationship with hydrogen production in the ultramafic samples. Sample C, for instance, has the highest Cr content but one of the lowest hydrogen yields, while Samples B and D, with lower Cr content, show a slight increase in hydrogen generation.

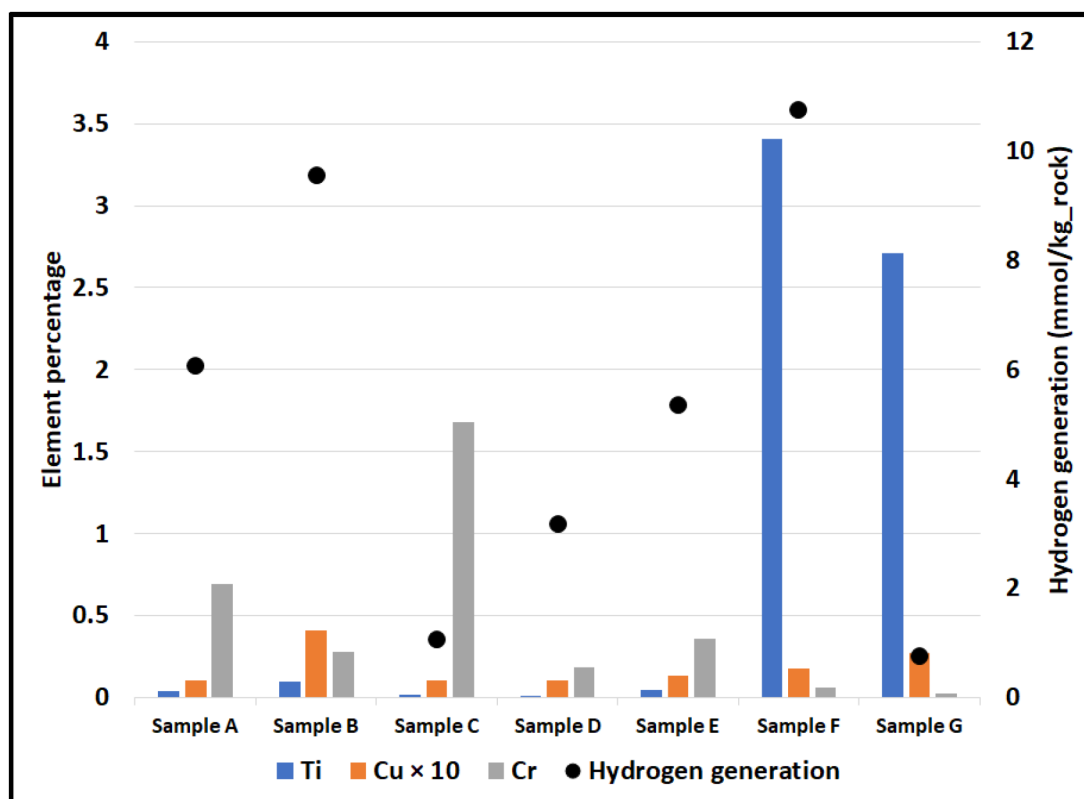


Figure 8—Plot of hydrogen generation and XRF percentage of three trace elements: titanium (Ti), copper (Cu), and chromium (Cr) across the seven rock samples.

In contrast, the mafic rocks (Samples F and G) exhibit a different behavior. Although both samples show elevated Ti levels, particularly Sample F, only Sample F generates a high amount of hydrogen, while Sample G produces very little despite also having high Ti. This suggests that Ti alone may not be sufficient to control hydrogen yield in mafic rocks. Cu displays an inverse relationship in this group, with Sample G having higher Cu and lower hydrogen generation, while Sample F shows the opposite. Although Cr content is small for the mafic rock group, sample F, with a larger chromium content, results in greater hydrogen generation. These contrasting trends between ultramafic and mafic rocks highlight that the influence of trace elements on hydrogen generation is likely rock-type dependent, with mineralogical context and broader geochemical interactions playing a vital role.

Conclusions

This study investigated the generation of hydrothermal hydrogen (H_2) across seven ultramafic and mafic rock types with varying mineralogies, aiming to elucidate the key reaction mechanisms and geochemical controls. The results revealed that differences in mineral composition and secondary mineral formation have a significant influence on hydrogen yield and reaction behavior.

Contrary to conventional assumptions, the study found that H_2 production in these systems does not always involve the formation of serpentine, brucite, or magnetite. Instead, the evidence suggests that surface-mediated redox reactions and the influence of trace mineral phases are at play. In some cases, minerals such as birnessite, andesine, and anorthite (in the case of mafic rocks) form instead of serpentine, indicating alternative Fe-driven pathways that challenge the conventional definition of serpentinization in such rock types.

Magnetite was often produced secondarily from olivine and did not undergo significant alteration, suggesting it acts as both a reaction product and a facilitator of hydrogen release. Continued H_2 production

appeared to rely on fluid-rock interactions, reactive surface renewal, and the presence of trace minerals in the system.

Fe-rich silicates such as olivine and pyroxene support sustained hydrogen generation through ongoing phase changes that renew reactive surfaces. In addition, aluminosilicates and transition metal-bearing minerals may enhance yields through geocatalytic effects, facilitating electron transfer or altering reaction kinetics. These findings highlight that hydrogen production is not governed solely by Fe oxidation, but by a combination of mineralogy, surface chemistry, and the geocatalytic influence of trace elements.

The findings expand the understanding of natural H₂ generation by emphasizing a broader interactions framework, one where mineral surface properties and trace minerals are central. This shift in perspective may improve hydrogen exploration strategies and inform the design of engineered systems for sustainable H₂ production from geologic sources. Future research should investigate reactions over extended durations, with sensitivity analyses across a range of temperatures, pressures, pH levels, and fluid compositions, including salinity. These studies should also account for hydrogen losses resulting from unintended migration and microbial activity.

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